

Approximation of Closure Operators for Chaotic Dynamical Systems Using Neural Networks

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Problem Formulation:

Consider a chaotic dynamical system combining slow and fast variables:

$$\mathbf{x}' = f(\mathbf{x})$$

The dynamics of a slow target variable x_1 are governed by:

$$x_1' = h(x_1, x_2, \dots, x_n)$$

Goal: Finding a Closure Operator C

- We aim to isolate x_1 from the rest of the system by approximating an operator C such that:

$$x_1' = h(x_1, C(x_1))$$

- This decouples x_1 to allow independent simulation and analysis.

L80 System Structure

- Represented by 9 variables, three 3D vectors: $\mathbf{x}, \mathbf{y}, \mathbf{z}$.
- **Target variable of interest:** Vector $\mathbf{y}$ (slow component, Rossby waves)
- Large enough Rossby number Ro induces strong interaction between slow Rossby waves and fast inertia-gravity waves (HLF Regime)
- This interaction of frequencies makes L80 an ideal benchmark.

Two-Step Parameterization Strategy

- Predict $\mathbf{z}$ from $\mathbf{y}$:

$$\mathbf{z} = \text{MLP}_1(\mathbf{y})$$

- Predict $\mathbf{x}$ from $\mathbf{y}$ and predicted $\mathbf{z}$:

$$\mathbf{x} = \text{MLP}_2(\mathbf{y}, \text{MLP}_1(\mathbf{y}))$$

- Small architectures: 1 hidden layer with 5 neurons (tanh activation).
- Trained using quadratic L_2 loss over 700 days of data (500 epochs).

Sampling Strategies

- NN_1 : Random sampling across simulation time.
- NN_2 : Predefined selection of continuous sample blocks.

Main Reconstruction Finding

- Despite achieving small offline L_2 loss values, **neither network achieved a satisfactory online reconstruction.**
- **Highlight:** NN_1 captured low-frequency trends significantly better than NN_2

Why did it fail?

1. Rare Events Phenomena

- HLF regime introduces rare attractor behaviors (e.g., attractor sojourns extending > 100 days).
- These rare events trick offline training, causing severe trajectory divergence online.

2. Spectral Bias

- Neural networks prioritize fitting low frequencies first.
- They capture to the general trend but miss the fast dynamics (I think this is due to the L2 norm, and can be solved on the training phase).

Possible Solution 1: Incorporating Lags / Memory

By Takens' embedding theorem, for a sufficiently large number of lags d , there exists a smooth map T such that:

$$x_2(k) = T\left(x_1(k), x_1(k-1), \dots, x_1(k-d)\right)$$

This introduces higher-order lags/derivatives to close the system for x_1 and aligns with the Mori-Zwanzig formulation (memory/convolution term).

Paper's Warning

- It might not be strictly a memory problem: classical Balance Equations (BE) manifolds reconstruct dynamics **without using memory**, but using a stochastic oscillator

Possible Solution 2: Neural SDE

Idea: I'm not sure about if this map C actually exists

If the map C doesn't exist, there is no deterministic closure!

Instead, we can use the NN to learn a map $\mathcal{G}$ and $\mathcal{B}$ such that it represents the dynamics ruled by h .

$$dx_1 = \mathcal{G}_\theta(x_1)dt + \mathcal{B}_\phi(x_1)dW_t$$

- The stochastic term compensates the lack of memory and mimics the feedback of the fast oscillations, which allows to continue with a first order system
- For the L80 system this should work as it they were already able to close the system using the BE benchmark + stochastic oscillators.

Possible Solution 3: Loss Functions & Architecture Improvements

Idea: The L_2 loss induces the spectral bias due to Parseval's theorem. Using H^1 norm penalizes derivative errors and forces the network to fit fast frequencies:

$$\mathcal{L}_{H^1} = \|y - \hat{y}\|_{L_2}^2 + \lambda \|\nabla y - \nabla \hat{y}\|_{L_2}^2$$

Architecture Improvements

- **Joint Training Architecture:** Train $\text{MLP}_1(\mathbf{z})$ and $\text{MLP}_2(\mathbf{x})$ simultaneously rather than sequentially.
- **Advanced Architectures:** Use an advanced architecture like Fourier Neural Operator, Wavelet Neural Operator to assess the High-Low frequency interaction
- **Physics-Informed Loss Terms:** Incorporate system physics or energy invariants directly into the loss function.

Discussion